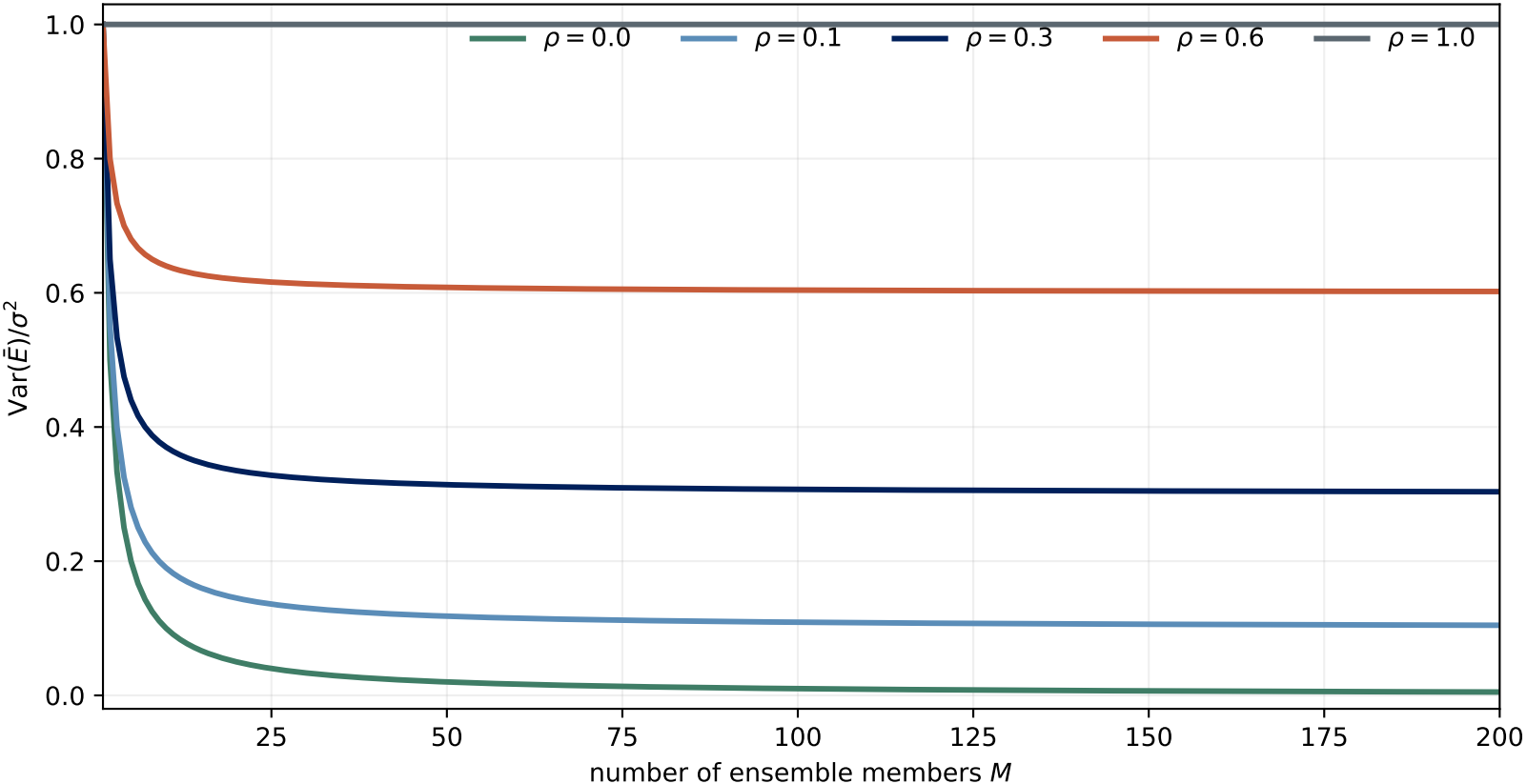
Averaging removes independent variation, not shared error
$\rho = 0.0$
$\rho = 0.1$
$\rho = 0.3$
$\rho = 0.6$
$\rho = 1.0$
$\mathrm{Var}(\bar{E})/\sigma^2$
1.0
0.8
0.6
0.4
0.2
0.0
25
50
75
100
125
150
175
200
number of ensemble members M